

EX NO:1 Butterworth IIR Filters

1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

PROGRAM

```
clc; clear; close all
```

% Given specifications

```
Rp = 1;    % Passband Ripple (dB)
```

```
Rs = 20;   % Stopband Attenuation (dB)
```

```
Fp = 1000; % Passband Frequency (Hz)
```

```
Fst = 3000; % Stopband Frequency (Hz)
```

```
Fs = 8000; % Sampling Frequency (Hz)
```

% Normalized frequencies

```
Wp = Fp / (Fs/2);
```

```
Ws = Fst / (Fs/2);
```

% Determine minimum filter order and cutoff frequency

```
[N, Wn] = buttord(Wp, Ws, Rp, Rs);
```

% Design Butterworth Low-Pass Filter

```
[b, a] = butter(N, Wn, 'low');
```

% Frequency response

```
figure;
```

```
freqz(b, a);
```

```
title('Frequency Response of Butterworth LPF');
```

% Impulse response

```
figure;
```

```
[h, n] = impz(b, a, 50);
```

```
stem(n, h);
```

```
title('Impulse Response');
```

% Step response

```
figure;
```

```

stepz(b, a);

title('Step Response');

% Pole-zero plot

figure;

zplane(b, a);

title('Pole-Zero Plot');

% Display results

fprintf('Minimum Filter Order (N) = %d\n', N);

fprintf('Cutoff Frequency (Wn) = %.4f\n', Wn);

```

OUTPUT

Minimum Filter Order (N) = 2
Cutoff Frequency (Wn) = 0.4159



